Introduction to Scrum

Module Objectives

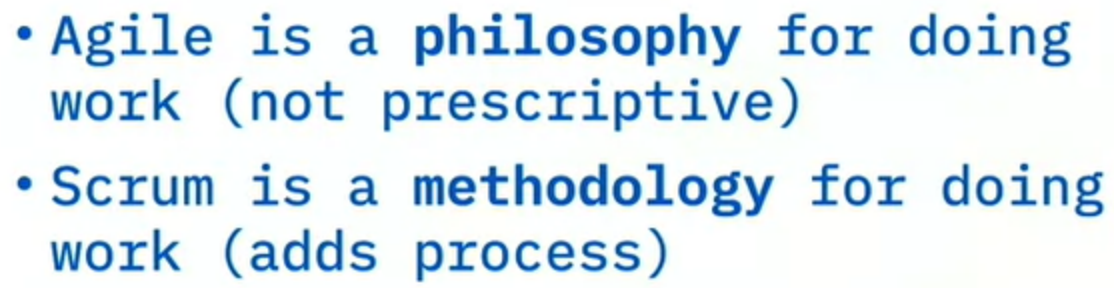
By the end of this module, you will be able to:

1. Define Agile and Scrum and distinguish between them.
2. Explain the key elements of the Scrum framework, including roles, artifacts, events, and timeboxes.
3. Describe the steps involved in the Scrum process, from backlog management to sprint execution and review.
4. Understand the iterative nature of Scrum and its focus on delivering value early and often.
5. Discuss the benefits of utilizing Scrum in software development projects, including increased productivity, enhanced product quality, reduced time to market, and improved stakeholder satisfaction.
6. Compare and contrast Scrum with Kanban, highlighting differences in cadence, release methodologies, roles, key metrics, and change philosophy

Scrum is an Agile project management framework that empowers teams to deliver valuable software in short, iterative cycles called sprints. Teams work collaboratively toward a shared goal, focusing on completing prioritized work items from a product backlog in each sprint. Regular meetings and transparent communication facilitate continuous improvement and adaptation, resulting in increased productivity, improved quality, and enhanced stakeholder satisfaction.

Agile vs. Scrum

Imagine **Agile** as the "why" and **Scrum** as the "how." Agile is a broader philosophy that emphasizes flexibility, customer collaboration, and delivering value early and often. Scrum, on the other hand, provides a specific framework for implementing these Agile principles. It outlines roles, artifacts, events, and timeboxes to structure the development process.



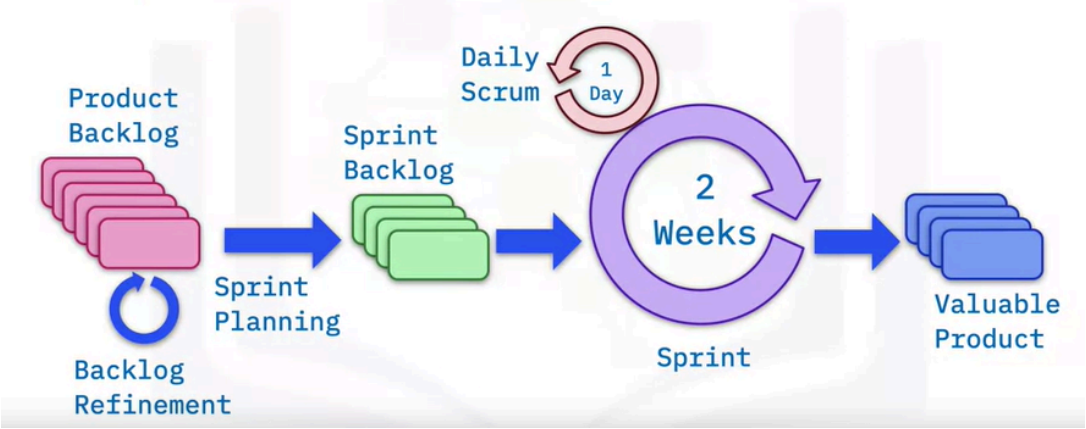
Key Characteristics of Scrum

- **Cross-functional teams:** Scrum teams consist of individuals with diverse skill sets who can work together independently without relying heavily on external expertise.
- **Self-organization:** Teams are empowered to make their own decisions regarding how to accomplish their goals within the Scrum framework.
- **Iterative development:** Scrum promotes breaking down work into smaller, time-boxed iterations called **sprints**. Each sprint focuses on delivering a potentially shippable product increment, allowing for continuous improvement and adaptation based on feedback.

The Scrum Process

Scrum follows a specific process flow:

- **Product Backlog:** This is the prioritized list of all features and functionalities the product should have. It serves as the single source of truth for the development team.
- **Backlog Refinement:** The team regularly reviews and refines the product backlog to ensure clarity and feasibility of user stories.
- **Sprint Planning:** The team collaborates during a sprint planning meeting to select user stories from the product backlog and commit to delivering them within the upcoming sprint.



- **Daily Scrum:** This short daily meeting (often standing up) allows team members to share progress, identify roadblocks, and adjust their work if needed. Answers three questions:
    1. What did you do yesterday/ What was done/completed yesterday?
    2. What will you do today?
    3. What (if anything) is blocking your progress?

- **Sprint Review:** At the end of a sprint, the team showcases the completed work to stakeholders and gathers feedback.
- **Sprint Retrospective:** This meeting helps the team reflect on the past sprint, identify areas for improvement, and adapt their approach for future sprints.

Scrum Roles

Product Owner:

- **Visionary and Decision-Maker:** Represents stakeholder interests and articulates the product vision.
- **Prioritization Guru:** Manages the product backlog, prioritizing features and requirements.
- **Acceptance Authority:** Decides whether to ship product increments based on their alignment with stakeholder needs.

Scrum Master:

- **Team Coach and Facilitator:** Guides the team through the Scrum process and removes roadblocks.

- **Agile Champion:** Promotes an environment that fosters self-organization and collaboration.
- **Process Protector:** Ensures adherence to Scrum practices and protects the team from external distractions.

Scrum Team:

- **Cross-Functional Powerhouse:** Comprises developers, testers, analysts, and other specialists working together.
- **Self-Organizing and Collaborative:** Decides work allocation and execution with minimal external influence.
- **Committed and Dedicated:** Negotiates commitments with the product owner for each sprint and focuses solely on delivering those goals.

Artifacts, Events, and Benefits

Scrum Artifacts

- **Product Backlog:** A prioritized list of features and functionalities for the product. It encompasses all future possibilities for the product.
- **Sprint Backlog:** A subset of the product backlog selected for completion within a specific **sprint** (a time-boxed iteration).
- **Done Increment:** A potentially shippable product increment developed and completed by the end of a sprint.

Scrum Events

- **Sprint Planning:** A collaborative meeting where the team plans the upcoming sprint, selecting items from the product backlog and committing to completing them.
- **Daily Scrum:** A short daily meeting (often called stand-up) for team members to discuss progress, identify roadblocks, and adjust their work if needed.
- **Sprint:** A time-boxed iteration (usually two weeks) where the team works on the selected sprint backlog items.
- **Sprint Review:** A meeting at the end of a sprint to showcase the completed work to stakeholders and gather feedback.
- **Sprint Retrospective:** A meeting to reflect on the past sprint, identifying areas for improvement and adapting the approach for future sprints.

Benefits of Scrum

- **Increased Productivity:** Regular meetings and transparency foster collaboration and self-management, leading to efficient work processes.
- **Improved Quality:** Continuous testing and engagement in development contribute to higher quality software.
- **Reduced Time to Market:** Working in short sprints allows for faster delivery of potentially functional product increments.
- **Enhanced Stakeholder Satisfaction:** Frequent interaction and early product demonstrations improve stakeholder engagement and satisfaction.
- **Stronger Team Dynamics:** Transparency and collaboration create a supportive and productive work environment.
- **Increased Employee Happiness:** Autonomy and control over work contribute to higher employee satisfaction.

Difference of Scrum and Kanban

	Scrum	Kanban
Cadence	Fixed length sprints	Continuous flow
Release Methodology	End of each sprint	Continuous delivery
Roles	Product Owner, Scrum Master, Development Team	No existing roles (optional Agile Coach)
Key metrics	Velocity	Cycle time
Change Philosophy	Teams should strive to not make changes to the sprint forecast during the sprint	Change can happen at any time

References

Sutherland, J., & Schwaber, K. (2017). *The Scrum Guide: The Definitive Guide to Scrum: The Rules of the Game*. Scrum.org.

Cohn, M. (2010). *Succeeding with Agile: Software Development Using Scrum*. Addison-Wesley Professional.